RELATIVE FINITE PRESENTATION IS INSENSITIVE TO A FINITE MAP OF FINITE TYPE ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted proof span contains an xymatrix commutative square, delivered verbatim; the preamble file already provides the xymatrix package, so no package was added. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- The definition of a module finitely presented relative to a base ring, and its behaviour under finite finitely presented ring maps. invoked at proof lines 21002-21005. Converts the exhibited presentation into the stated equivalence.
- Finiteness of A -> B as an A-module, giving monic equations for the generators. used at proof lines 20991-20993. Produces the finitely many relations presenting B over a polynomial ring.
- The choice of a surjection from a polynomial ring onto a finite type algebra. used at proof lines 20989-20990. Provides the left column of the commutative square.

Source reference. Original lemma 0564: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6697–6697. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let R be a ring. Let $A \to B$ be a finite map of finite type R-algebras. Let M be a B-module. Then M is an A-module finitely presented relative to R if and only if M is a B-module finitely presented relative to R.*

Proof. Choose a surjection $R[x_1, \ldots, x_n] \to A$. Choose $y_1, \ldots, y_m \in B$ which generate B over A. As $A \to B$ is finite each y_i satisfies a monic equation with coefficients in A. Hence we can find monic polynomials $P_j(T) \in R[x_1, \ldots, x_n][T]$ such

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that $P_j(y_j) = 0$ in B. Then we get a commutative diagram

$$\begin{array}{ccc} R[x_1, \ldots, x_n] & \longrightarrow & R[x_1, \ldots, x_n, y_1, \ldots, y_m]/(P_j(y_j)) \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array}$$

Since the top arrow is a finite and finitely presented ring map we conclude by Algebra, Lemma 0564 and the definition. $\square$